Ramsey–Cass–Koopmans Model

Lecture 2 Cheatsheet — 14E022 Macroeconomics I

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Contents

1 Notation and Definitions 2

2 Environment and Welfare Theorems 2

3 Household Problem 3

4 Hamiltonian Derivation 3

5 CRRA Utility and the Euler Equation 4

6 Firm Problem 5

7 Dynamic System 5

8 Steady State and Modified Golden Rule 6

9 Phase Diagram 7

10 Comparative Statics 8

11 Comparison to Solow Model 8

12 Data Application and Savings Rate Puzzle 9

13 Problem-Solving Templates 9

14 Common Pitfalls 10

15 Quick Reference Formulas 11

1 Notation and Definitions

Symbol	Definition	Units / Notes
K_t	Aggregate capital stock	Level
$L_t = N_t$	Labor / population	Level
C_t, c_t	Aggregate / per-worker consumption	$c_t = C_t/L_t$
k_t	Per-worker capital	$k_t = K_t/L_t$
$\hat{k}_t$	Per-effective-worker capital	$\hat{k}_t = K_t/(B_t L_t)$
$\hat{c}_t$	Per-effective-worker consumption	$\hat{c}_t = C_t/(B_t L_t)$
$\hat{y}_t$	Per-effective-worker output	$\hat{y}_t = f(\hat{k}_t)$
a_t	Per-worker assets (= k_t in eq.)	
n	Population growth rate	$\dot{N}/N = n$
x	Rate of labor-augmenting tech. progress	$\dot{B}/B = x$
δ	Capital depreciation rate	
ρ	Household rate of time preference	$\rho > 0$
σ	Inverse elasticity of intertemporal sub.	$\sigma > 0$
α	Capital share in Cobb-Douglas	$0 < \alpha < 1$
ν_t	Current-value co-state (shadow price)	
r_t	Rental rate of capital	$r_t = f'(k_t) - \delta$
w_t	Wage	$w_t = f(k_t) - k_t f'(k_t)$

Key Transformations

$$\hat{k}_t = \frac{k_t}{B_t}, \qquad \hat{c}_t = \frac{c_t}{B_t}, \qquad \hat{y}_t = \frac{y_t}{B_t} = f(\hat{k}_t).$$

Growth rates obey $\gamma_x \equiv \dot{x}/x$. For a hat variable $\hat{z} = z/B$:

$$\gamma_{\hat{z}} = \gamma_z - x.$$

2 Environment and Welfare Theorems

Demographics and Technology

$$N_t = N_0\, e^{nt}, \qquad B_t = B_0\, e^{xt} \quad \text{(labor-augmenting; Uzawa theorem).}$$

Effective labor: $A_t = B_t N_t$.

Preferences

The representative household has $N_0 e^{nt}$ members and maximizes

$$U_0 = \int_0^\infty u(c_t)\, e^{-(\rho-n)t}\, dt,$$

where $u(c_t)$ is instantaneous utility, $\rho - n > 0$ is required for convergence.

Welfare Theorems

Under the standard assumptions (no externalities, complete markets, no distortions), the **First and Second Welfare Theorems** apply:

- The competitive equilibrium allocation is Pareto optimal.
- The social planner's problem and the decentralized equilibrium yield *identical* allocations.
- Consequence: we can solve either problem and obtain the same dynamics.

3 Household Problem

Budget Constraint

In per-worker terms, assets a_t evolve as:

$$\dot{a}_t = w_t + r_t a_t - c_t - n a_t,$$

or equivalently,

$$\dot{a}_t + n a_t = w_t + r_t a_t - c_t. \tag{BC}$$

In market clearing, $a_t = k_t$.

No-Ponzi Condition

$$\lim_{t\to\infty} a_t \exp\left(-\int_0^t \big(r(s) - n\big)\, ds\right) \geq 0. \tag{NPC}$$

This prevents households from rolling over debt forever (dying in debt). Combined with the optimality condition (TVC), this holds with equality in the optimum.

Transversality Condition (TVC)

$$\lim_{t\to\infty} \nu_t\, a_t = 0. \tag{TVC}$$

where ν_t is the shadow price of assets in the current-value Hamiltonian. The TVC ensures the household does not leave unconsumed wealth at infinity.

4 Hamiltonian Derivation

Setup

- **Control variable:** c_t (consumption).
- **State variable:** a_t (assets / capital).
- **Co-state variable:** ν_t (shadow price of assets).

Present-Value Hamiltonian

$$\mathcal{J}(c_t,\, a_t,\, \nu_t,\, t) = u(c_t)\, e^{-(\rho-n)t} + \nu_t \Big[w_t + (r_t - n)\, a_t - c_t\Big]. \tag{PVH}$$

FOC 1 — control variable:

$$\frac{\partial \mathcal{J}}{\partial c_t} = 0 \implies \nu_t = u_c(c_t)\, e^{-(\rho-n)t}. \tag{F1}$$

FOC 2 — co-state equation:

$$\frac{\partial \mathcal{J}}{\partial a_t} = -\dot{\nu}_t \implies \dot{\nu}_t = -(r_t - n)\, \nu_t. \tag{F2}$$

FOC 3 — state equation (reproduced):

$$\frac{\partial \mathcal{J}}{\partial \nu_t} = \dot{a}_t \implies \dot{a}_t = w_t + (r_t - n)\, a_t - c_t. \tag{F3}$$

Euler equation from F1 and F2:

Take logs of (F1): $\ln \nu_t = \ln u_c(c_t) - (\rho - n)t$.
Differentiate with respect to t:

$$\frac{\dot{\nu}_t}{\nu_t} = \frac{u_{cc}(c_t)\, \dot{c}_t}{u_c(c_t)} - (\rho - n).$$

Substitute (F2) for $\dot{\nu}_t/\nu_t = -(r_t - n)$:

$$-(r_t - n) = \frac{u_{cc}(c_t)\, \dot{c}_t}{u_c(c_t)} - (\rho - n).$$

Rearrange:

$$\frac{\dot{c}_t}{c_t} = -\frac{u_c(c_t)}{u_{cc}(c_t)\, c_t}\, (r_t - \rho).$$

Define the **elasticity of intertemporal substitution** $\varepsilon(c) \equiv -u_c/(u_{cc}c)$:

$$\boxed{\frac{\dot{c}_t}{c_t} = \varepsilon(c_t)\, (r_t - \rho).}$$

5 CRRA Utility and the Euler Equation

CRRA Functional Form

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \qquad \sigma > 0,\ \sigma \neq 1.$$

For $\sigma = 1$: $u(c) = \ln c$ (log utility).

Properties: $u_c = c^{-\sigma}$, $u_{cc} = -\sigma c^{-\sigma-1}$, so

$$\varepsilon(c) = -\frac{u_c}{u_{cc}\, c} = \frac{1}{\sigma}.$$

Euler Equation (CRRA)

$$\frac{\dot{c}_t}{c_t} = \frac{1}{\sigma}(r_t - \rho).$$

Interpretation: consumption grows fast when the return to saving r_t exceeds the discount rate ρ. σ governs how strongly households respond to interest rate incentives.

In Per-Effective-Worker Terms

Since $\hat{c} = c/B$ and $\dot{B}/B = x$:

$$\frac{\dot{\hat{c}}}{\hat{c}} = \frac{\dot{c}}{c} - x = \frac{1}{\sigma}(r_t - \rho) - x.$$

Using $r_t = f'(\hat{k}) - \delta$:

$$\frac{\dot{\hat{c}}}{\hat{c}} = \frac{1}{\sigma}\big[f'(\hat{k}_t) - \delta - \rho - \sigma x\big]. \tag{Euler-hat}$$

6 Firm Problem

Technology

Firms have CRS production: $Y_t = F(K_t, B_t L_t)$, or in intensive form $\hat{y} = f(\hat{k})$ where f is concave, $f(0) = 0$, Inada conditions hold.

Profit Maximization (competitive firms)

Firms rent capital and hire labor each period:

$$\max_{K,L} \; F(K, BL) - r_t K - w_t L.$$

FOCs (using Euler's theorem for CRS):

$$r_t = F_K = f'(\hat{k}_t), \qquad w_t = F_{BL} \cdot B_t = B_t\big[f(\hat{k}_t) - \hat{k}_t f'(\hat{k}_t)\big].$$

Note: in equilibrium $r_t = f'(\hat{k}) - \delta$ (net of depreciation).

Cobb-Douglas Special Case

$$f(\hat{k}) = \hat{k}^\alpha, \quad f'(\hat{k}) = \alpha\hat{k}^{\alpha-1}, \quad f(\hat{k}) - \hat{k}f'(\hat{k}) = (1-\alpha)\hat{k}^\alpha.$$

7 Dynamic System

The equilibrium is fully described by two ODEs in $(\hat{k}_t, \hat{c}_t)$:

$$\dot{\hat{k}}_t = f(\hat{k}_t) - \hat{c}_t - (n + x + \delta)\,\hat{k}_t, \tag{K-dot}$$

$$\frac{\dot{\hat{c}}_t}{\hat{c}_t} = \frac{1}{\sigma}\big[f'(\hat{k}_t) - \delta - \rho - \sigma x\big]. \tag{C-dot}$$

Derivation of (K-dot)

The aggregate resource constraint is $\dot{K} = Y - C - \delta K$. Divide by BL, use $(\dot{K}/(BL)) = \dot{\hat{k}} + (n+x)\hat{k}$:

$$\dot{\hat{k}} = f(\hat{k}) - \hat{c} - (n + x + \delta)\hat{k}.$$

Boundary Conditions

- Initial condition: $\hat{k}_0$ given (predetermined state).
- TVC / No-Ponzi at infinity: pins down the initial value of $\hat{c}_0$ (the *jump* variable).

8 Steady State and Modified Golden Rule

Conditions

Set $\dot{\hat{k}} = 0$ and $\dot{\hat{c}} = 0$:

$$\dot{\hat{c}}^* = 0 \implies \boxed{f'(\hat{k}^*) = \delta + \rho + \sigma x} \qquad \text{(Modified Golden Rule)} \qquad \text{(MGR)}$$

$$\dot{\hat{k}}^* = 0 \implies \hat{c}^* = f(\hat{k}^*) - (n + x + \delta)\hat{k}^* \qquad \text{(resource balance)}$$

The Modified Golden Rule (MGR) pins down $\hat{k}^*$ *independently of* n, unlike the Solow model.

Cobb-Douglas Closed Forms

With $f(\hat{k}) = \hat{k}^\alpha$, from MGR: $\alpha(\hat{k}^*)^{\alpha-1} = \delta + \rho + \sigma x$, so

$$\hat{k}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{1/(1-\alpha)}, \qquad \hat{y}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{\alpha/(1-\alpha)}.$$

Steady-State Interest Rate and Savings Rate

$$r^* = f'(\hat{k}^*) - \delta = \rho + \sigma x.$$

$$s^* = \frac{(n + x + \delta)\hat{k}^*}{f(\hat{k}^*)} = \frac{\alpha(n + x + \delta)}{\delta + \rho + \sigma x + n}.$$

With Cobb-Douglas: $s^* = \frac{\alpha(n+x+\delta)}{\delta+\rho+\sigma x} \cdot \frac{\hat{k}^*}{yhat^*} \cdot \frac{1}{\hat{k}^*} \approx \frac{\alpha(n+x+\delta)}{\rho+\sigma x+\delta}$, which is *less than* the Solow savings rate $s = \alpha$ at the Golden Rule.

TVC Implication: No Dynamic Inefficiency

The TVC requires:

$$n + (1 - \sigma)x - \rho < 0.$$

This ensures $\hat{k}^* < \hat{k}^*_{\text{Golden Rule}}$, i.e., the economy does *not* over-accumulate capital. Dynamic inefficiency is ruled out by assumption; the Ramsey model satisfies the Modified (not the standard) Golden Rule.

Recall: Golden Rule maximizes $\hat{c}$, requiring $f'(\hat{k}_g) = n + x + \delta$. Since $\rho + \sigma x > 0$, we have $f'(\hat{k}^*) > f'(\hat{k}_g)$, hence $\hat{k}^* < \hat{k}_g$.

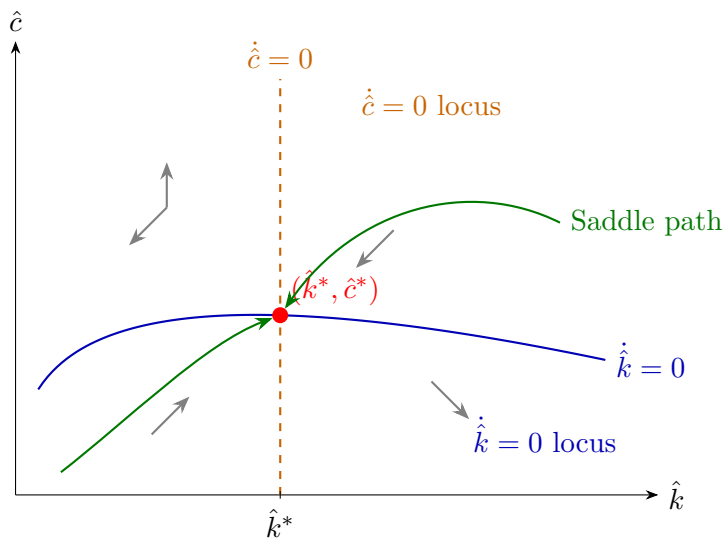
9 Phase Diagram

Loci and Dynamics

- $\dot{\hat{c}} = 0$: vertical line at $\hat{k} = \hat{k}^*$ (from MGR).
- $\dot{\hat{k}} = 0$: inverted-U curve $\hat{c} = f(\hat{k}) - (n + x + \delta)\hat{k}$; peaks at Golden Rule $\hat{k}_g$ where $f'(\hat{k}_g) = n + x + \delta$.
- Above $\dot{\hat{k}} = 0$: $\dot{\hat{k}} < 0$ (capital declines). Below: $\dot{\hat{k}} > 0$.
- Right of $\dot{\hat{c}} = 0$: $\dot{\hat{c}} < 0$ (consumption declines). Left: $\dot{\hat{c}} > 0$.

Saddle-Path Stability

The steady state is a **saddle point**: one stable and one unstable manifold. The **saddle path** (stable arm) is the unique trajectory converging to $(\hat{k}^*, \hat{c}^*)$. The initial value $\hat{c}_0$ is chosen to place the economy on the saddle path.



10 Comparative Statics

Parameter	Δ	Affected locus	$\hat{k}^*$	$\hat{c}^*$
ρ (impatience) $\uparrow$	$+$	$\dot{\hat{c}} = 0$ shifts left	$\downarrow$	$\downarrow$
σ (risk aversion) $\uparrow$	$+$	$\dot{\hat{c}} = 0$ shifts left (if $x > 0$)	$\downarrow$	$\downarrow$
δ (depreciation) $\uparrow$	$+$	$\dot{\hat{c}} = 0$ shifts left; $\dot{\hat{k}} = 0$ shifts down	$\downarrow$	$\downarrow$
n (pop. growth) $\uparrow$	$+$	$\dot{\hat{k}} = 0$ shifts down	*none*	$\downarrow$
x (tech. progress) $\uparrow$	$+$	$\dot{\hat{c}} = 0$ shifts left; $\dot{\hat{k}} = 0$ shifts down	$\downarrow$	—
α (capital share) $\uparrow$	$+$	Both loci shift	$\uparrow$	$\uparrow$

Effect of an Increase in ρ

1. $\dot{\hat{c}} = 0$ locus shifts *left* (households want to consume more now).
2. Instantaneous jump up in $\hat{c}_0$ (control variable can jump).
3. $\hat{k}$ falls over transition (more consumption, less investment).
4. New steady state: lower $\hat{k}^*$ and $\hat{c}^*$.

Key: *n does not affect $\hat{k}^*$*, unlike the Solow model.

Effect of n on Steady State

From MGR: $f'(\hat{k}^*) = \delta + \rho + \sigma x$ — no n appears! So $\hat{k}^*$ is independent of n. However, $\hat{c}^* = f(\hat{k}^*) - (n + x + \delta)\hat{k}^*$ falls as n rises (more workers dilute capital).

11 Comparison to Solow Model

Feature	Solow	Ramsey
Savings rate s	Exogenous constant	Endogenous (optimized)
Steady state $\hat{k}^*$	$f(\hat{k}^*) = (n + x + \delta)\hat{k}^*/s$	$f'(\hat{k}^*) = \delta + \rho + \sigma x$
Effect of n on $\hat{k}^*$	$\hat{k}^* \downarrow$ as $n \uparrow$	None (no n in MGR)
Dynamic inefficiency	Possible ($\hat{k} > \hat{k}_g$)	Ruled out by TVC
Welfare optimality	Not guaranteed	Yes (Welfare Theorems)
Transition dynamics	Exogenous	Driven by Euler equation
Savings in transition	Constant	Varies (declining toward SS)

Why Ramsey Cannot Be Dynamically Inefficient

In Solow, if $s > s_g = \alpha$ (Cobb-Douglas), the economy over-accumulates. In Ramsey the TVC forces:

$$n + (1 - \sigma)x < \rho \implies f'(\hat{k}^*) = \delta + \rho + \sigma x > \delta + n + x = f'(\hat{k}_g).$$

So $\hat{k}^* < \hat{k}_g$ always. Households would never save so much that $r < n + x$ in steady state.

12 Data Application and Savings Rate Puzzle

Calibration Example

Assume $x = 0.02$, $n = 0.01$, $\alpha = 0.33$, $k/y \approx 3$, $i/y \approx 0.25$. Then:

$$\delta = \frac{i/y}{k/y} - (n + x) = \frac{0.25}{3} - 0.03 \approx 0.053.$$

From steady state $r^* = \rho + \sigma x$ and observed $r^* \approx 0.077$:

$$\rho + \sigma x \approx 0.077 - 0.053 = 0.024.$$

(Typical values: $\rho \approx 0.02$, $\sigma = 2$, giving $\rho + \sigma x = 0.06$; calibrations vary.)

The Savings Rate Puzzle

Model predictions along the transition:

- If $\hat{k}_0 < \hat{k}^*$: high r_t, so from Euler $\dot{\hat{c}}/\hat{c} > 0$ (consumption growing fast).
- Investment share (i/y) can be non-monotone; model generates *declining* saving rates once near SS.
- **Puzzle:** many developing economies (Japan 1960–2000, China) show *rising then falling* (hump-shaped) savings rates. The Ramsey model only matches the declining part.
- Possible resolutions: subsistence consumption, demographic effects (life-cycle), financial development removing liquidity constraints.

Country Examples

- **Japan 1960–2000:** savings rate peaked around 1975, then fell as Japan approached steady state. Consistent with the transitional decline predicted by Ramsey.
- **China 2000–present:** savings rate still rising, inconsistent with simple Ramsey (far from SS, or other mechanisms at work).

13 Problem-Solving Templates

Template A: Setting Up the Hamiltonian

1. Write the objective: $\int_0^\infty u(c_t)e^{-(\rho-n)t}\,dt$.
2. Write the state equation (constraint on $\dot{a}$ or $\dot{\hat{k}}$).
3. Form the **present-value Hamiltonian**: $\mathcal{J} = [\text{integrand}] + \nu_t \cdot [\text{state equation RHS}]$.
4. Derive FOCs: $\mathcal{J}_c = 0$, $\mathcal{J}_a = -\dot{\nu}_t$, $\mathcal{J}_\nu = \dot{a}_t$.
5. Write TVC: $\lim_{t\to\infty} \nu_t a_t = 0$.
6. Eliminate ν_t to get Euler equation.

Template B: Finding the Steady State

1. Set $\dot{\hat{c}} = 0$ in (C-dot): solve $f'(\hat{k}^*) = \delta + \rho + \sigma x$.
2. Set $\dot{\hat{k}} = 0$ in (K-dot): solve $\hat{c}^* = f(\hat{k}^*) - (n + x + \delta)\hat{k}^*$.
3. For Cobb-Douglas $f(\hat{k}) = \hat{k}^\alpha$: $\hat{k}^* = [\alpha/(\delta + \rho + \sigma x)]^{1/(1-\alpha)}$.
4. Verify $\hat{c}^* > 0$ (otherwise model breaks down).

Template C: Phase Diagram Analysis

1. Draw $\dot{\hat{k}} = 0$ locus (inverted-U peaking at Golden Rule $\hat{k}_g$).
2. Draw $\dot{\hat{c}} = 0$ locus (vertical line at $\hat{k}^*$; note $\hat{k}^* < \hat{k}_g$).
3. Mark steady state at intersection.
4. Determine signs: above $\dot{\hat{k}} = 0$: $\dot{\hat{k}} < 0$; right of $\dot{\hat{c}} = 0$: $\dot{\hat{c}} < 0$.
5. Draw saddle path (stable arm): downward sloping from upper-left and lower-right.
6. Initial condition: $\hat{k}_0$ given; $\hat{c}_0$ chosen on saddle path.

Template D: Comparative Statics

1. Identify which locus shifts (check which equation contains the parameter).
2. Determine direction of shift.
3. Find new intersection $(\hat{k}^{**}, \hat{c}^{**})$.
4. Describe **impact effect**: jump in $\hat{c}$ (instantaneous); $\hat{k}$ is predetermined.
5. Describe **transition**: path along new saddle path to new steady state.
6. Compute changes in r^*, w^*, s^* if required.

14 Common Pitfalls

1. **Wrong discount factor in objective.** The household objective uses $e^{-(\rho-n)t}$, *not* $e^{-\rho t}$. The n comes from the household size growing at rate n (dynasty normalization).
2. **Confusing current-value and present-value Hamiltonians.** In the present-value Hamiltonian, the co-state equation is $-\dot{\nu} = \mathcal{J}_a$. In the current-value Hamiltonian $\hat{\mathcal{H}} = e^{(\rho-n)t}\mathcal{J}$, the co-state equation is $-\dot{\mu} + (\rho - n)\mu = \hat{\mathcal{H}}_a$ (different!).
3. **Forgetting σx in the MGR.** The Euler equation in hat variables gives $f'(\hat{k}^*) = \delta + \rho + \sigma x$, *not* $\delta + \rho$. The σx term comes from converting per-worker to per-effective-worker growth rates.
4. **n affects $\hat{k}^*$?** No. Unlike Solow, n does not appear in the MGR. It only affects $\hat{c}^*$ and transition speed.
5. **Dynamic inefficiency in Ramsey.** It cannot occur. The TVC prevents over-accumulation. Do not apply the golden rule test ($r^* \lesseqgtr n + x$) to check for inefficiency; the Ramsey outcome always satisfies $r^* = \rho + \sigma x > 0$.

6. $\hat{k}_0$ **vs.** $\hat{c}_0$**.** Capital $\hat{k}$ is a *state* variable (predetermined, cannot jump). Consumption $\hat{c}$ is a *control* variable (can jump). At time $t = 0$, $\hat{k}_0$ is given; $\hat{c}_0$ is chosen to be on the saddle path.

7. **TVC sign.** The no-Ponzi condition is ≥ 0; optimality turns it into equality. At the optimum: $\lim_{t\to\infty} \nu_t a_t = 0$.

8. **Saddle path slope.** The saddle path is *upward* sloping in $(\hat{k}, \hat{c})$ space (not downward). Near the SS it has positive slope.

15 Quick Reference Formulas

$$\frac{\dot{c}_t}{c_t} = \varepsilon(c_t)\,(r_t - \rho), \qquad \varepsilon(c) = -\frac{u_c}{u_{cc}\,c}.$$

$$\frac{\dot{\hat{c}}}{\hat{c}} = \frac{1}{\sigma}\big[f'(\hat{k}) - \delta - \rho - \sigma x\big].$$

$$\dot{\hat{k}} = f(\hat{k}) - \hat{c} - (n + x + \delta)\hat{k}.$$

$$f'(\hat{k}^*) = \delta + \rho + \sigma x.$$

$$\hat{k}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{1/(1-\alpha)}, \qquad \hat{y}^* = \left(\frac{\alpha}{\delta + \rho + \sigma x}\right)^{\alpha/(1-\alpha)}.$$

$$r^* = \rho + \sigma x, \qquad s^* = \frac{(n + x + \delta)\hat{k}^*}{f(\hat{k}^*)}.$$

$$n + (1 - \sigma)x < \rho \implies \hat{k}^* < \hat{k}^*_g.$$

Present-Value Hamiltonian Summary

$$\mathcal{J} = u(c_t)\,e^{-(\rho-n)t} + \nu_t\big[w_t + (r_t - n)a_t - c_t\big].$$

$$
\begin{aligned}
\mathcal{J}_c = 0 &\implies \nu_t = u_c(c_t)\, e^{-(\rho-n)t}. \\
\mathcal{J}_a = -\dot{\nu}_t &\implies \dot{\nu}_t = -(r_t - n)\nu_t. \\
\mathcal{J}_\nu = \dot{a}_t &\implies \dot{a}_t = w_t + (r_t - n)a_t - c_t. \\
\text{TVC:} \quad & \lim_{t\to\infty} \nu_t a_t = 0.
\end{aligned}
$$